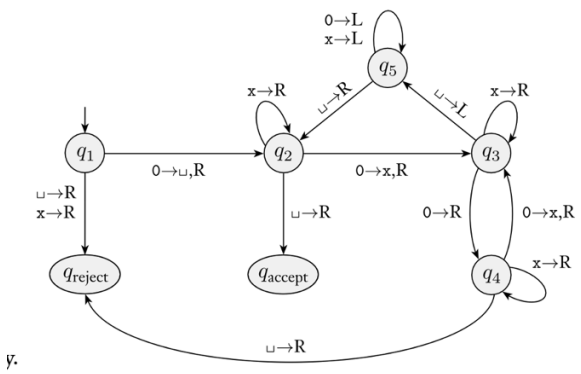
Lab 7

Submission by: **April 21st - 4pm**

1) [0.4 points]
Consider a 2-PDA that is a pushdown automaton equipped with two stacks, that can decide to push a symbol on neither stack, on either of the two stacks or on both stacks. Show how such a 2-PDA can recognise the language $A = \{a^k b^k c^k \mid k \geq 0\}$.

2) [0.2 points]
Consider the TM M seen at lecture that decides the language $B = \{0^{2^n} \mid n \geq 0\}$ with state diagram:



Give the sequence of configurations that the M enters when started on the following input strings:

a. 0
b. 00
c. 000
d. 000000

3) [0.2 points]

Examine the formal definition of a Turing machine to answer the following questions, and explain your reasoning.

a. Can a Turing machine ever write the blank symbol ⊔ on its tape?
b. Can the tape alphabet Γ be the same as the input alphabet Σ?
c. Can a Turing machine's head *ever* be in the same location in two successive steps?
d. Can a Turing machine contain just a single state?

4) [0.4 points]
Design the state diagram of the Turing Machine discussed at lecture that recognises the language $B = \{a^k b^k c^k \mid k \geq 0\}$

5) [0.3 points]
Give *implementation-level* descriptions of Turing machines that decide the following languages over the alphabet {0,1}.

a. $\{w \mid w$ contains an equal number of 0s and 1s$\}$
b. $\{w \mid w$ contains twice as many 0s as 1s$\}$
c. $\{w \mid w$ does not contain twice as many 0s as 1s$\}$

6) [0.5 points]

A ***Turing machine with left reset*** is similar to an ordinary Turing machine, but the transition function has the form

$$\delta\colon Q \times \Gamma \longrightarrow Q \times \Gamma \times \{\mathrm{R}, \mathrm{RESET}\}.$$

If $\delta(q, a) = (r, b, \mathrm{RESET})$, when the machine is in state q reading an a, the machine's head jumps to the left-hand end of the tape after it writes b on the tape and enters state r. Note that these machines do not have the usual ability to move the head one symbol left. Show that Turing machines with left reset recognize the class of Turing-recognizable languages.

7) [0.7 points]

A ***Turing machine with stay put instead of left*** is similar to an ordinary Turing machine, but the transition function has the form

$$\delta\colon Q \times \Gamma \longrightarrow Q \times \Gamma \times \{\mathrm{R}, \mathrm{S}\}.$$

At each point, the machine can move its head right or let it stay in the same position. Show that this Turing machine variant is *not* equivalent to the usual version. What class of languages do these machines recognize?

8) [0.25 points]
Show that the collection of decidable languages is closed under the operation of:

a. union.
b. concatenation.
c. star.
d. complementation.
e. intersection.

9) [0.15 points]

Show that the collection of Turing-recognisable languages is closed under the operation of:

a. union.

b. concatenation.

c. star.

10) [0.3 points]

Give Turing machine programs for the following operations:

a. Given a positive binary number x return $x - 1$

b. Given two binary numbers x and $y, x > y$ return $x - y$

c. Given two binary numbers x and $y, x > y$ return x/y

11) [0 points]

Show that single-tape TMs that cannot write on the portion of the tape containing the input string recognise only regular languages.